

James Irwin

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Security Clearance: Active Top Secret

Education

Cal Poly, San Luis Obispo | MS in Computer Science Expected: May 2027

- **Proposed Thesis:** *Understanding the Design and Impact of Digital Interfaces for Music Creation*

- **GPA:** 4.00/4.00

Cal Poly, San Luis Obispo | BS in Computer Science

Graduated: June 2026

- **Minors:** Data Science and Philosophy

- **GPA:** 3.76/4.00

- **Award:** Cal Poly Merit Scholar

Skills

Languages: C, C++, C, C++, ChucK, JavaScript, MATLAB, Python, R, SQL, TypeScript

Tools & Frameworks: Azure, Flask, Git, librosa, NumPy, pandas, scikit-learn, TensorFlow, tidymodels, Unity

Work and Research Experience

Software Engineer Intern | Northrop Grumman – San Diego, CA

June 2026 – Present

- Worked full-time during Summer 2026 and have continued part-time while completing master's degree.
- Developing an application to ingest, process, and visualize 3D flight data in real time.
- Designed, implemented, and tested an end-to-end system for transmitting RTK corrections via NTRIP between a base station and a moving receiver.

Machine Learning Research Assistant | Sound Ethics – Remote

Jan 2026 – June 2026

- Collaborated with Sound Ethics through Cal Poly's Data Science Capstone to research inference-time attribution and controllable AI music generation.
- Developed a multi-input music generation pipeline using **ACE-Step**, **Demucs**, and **librosa**, enabling conditioning on instrument components from up to **five reference tracks** without training a new generative model.
- Conducted a **14-participant blind user study** comparing our framework with baseline ACE-Step. Our framework received higher average ratings for sound quality, prompt adherence, input adherence, and cohesiveness.
- Presented research findings at the UCSB Data Science Capstone Showcase.

Software Engineer Intern | Speridian Technologies - Remote

June 2025 – Sept 2025

- Researched, implemented, and evaluated open-source alternatives to Azure AI Document Intelligence for automated document processing.
- Developed Python pipelines to transform OCR model output from PDFs into structured JSON.
- Built an embedding-based document classification approach using **Azure**, generating vector representations and measuring similarity between documents.

Projects

SoundLink

Winter 2026 - Spring 2026

- Developed a six-level music-based puzzle game in **Unity** where players arrange 2D puzzle pieces to construct rhythmic compositions through visual and auditory cues.
- Integrated **ChucK with Unity via Chunity** to synchronize audio timing with game events.
- Conducted playtesting and iterated on game controls based on user feedback.

Photographer Attribution with Neural Networks

Fall 2025

- Built a photographer-attribution neural network in **TensorFlow** using MobileNetV2 to classify images from a 505-photo dataset without relying on EXIF metadata.
- Engineered a feature pipeline combining RGB and HSV image embeddings, grayscale-derived composition features, and 20 hand-labeled subject/composition features.
- Used 5-fold stratified cross-validation and balanced accuracy for hyperparameter tuning, improving cross-validation performance from **79.4% to 87.8% balanced accuracy**.